Movie Rental Data Warehouse Design Report

High-Level Data Warehouse Design from an OLTP Movie Rental System

Course: Data Warehousing / Data Architecture
Source OLTP Database: Sakila
Target Data Warehouse: movie_rental_dw

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1. Introduction

This project presents the design and implementation of a data warehouse for a movie rental business using the Sakila OLTP database as the source system. The original OLTP database supports daily operational activities such as managing customers, films, inventory, rentals, payments, staff, stores, and locations. This type of database is suitable for transaction processing because it stores detailed operational records and supports frequent inserts, updates, and lookups.

However, the OLTP schema is not optimized for analytical reporting. Business managers need to answer questions about rental activity, revenue, customer behavior, film popularity, staff performance, store performance, inventory availability, and trends over time. For this reason, a separate data warehouse database named movie_rental_dw was created.

The data warehouse was designed using dimensional modeling. Instead of copying the OLTP schema directly, the data was reorganized into fact tables and dimension tables. The fact tables store measurable business events, while the dimension tables provide descriptive context for analysis. The final warehouse supports decision-making by making business reports easier and faster to generate.

The project was implemented using MySQL Workbench. The original Sakila SQL file was imported first, then a separate data warehouse schema was created. ETL SQL scripts were used to extract, transform, and load the data from the OLTP database into the warehouse. Analytical SQL reports were then written to answer the business questions.

2. Business Questions

Before designing the data warehouse, the main analytical needs of the movie rental business were identified. The warehouse was designed to answer the following business questions:

1. Which films are rented most frequently?
2. Which films generate the highest revenue?
3. Which stores generate the highest revenue?
4. How does revenue change by month?
5. How does rental activity change by month?
6. Which customers rent the most films and generate the highest revenue?
7. Which staff members process the highest number of rentals and payments?
8. Which films are returned late most often?

These questions are important because they help managers understand business performance from different perspectives. Film rental and revenue reports help identify the most valuable films. Store reports help compare branch performance. Customer reports help identify highly active customers. Staff performance reports support operational evaluation. Monthly trend reports help managers understand changes in revenue and rental activity over time. Late return reports can also help the business improve rental policies and inventory planning.

3. Dimensional Model Design

The proposed warehouse design is mainly a star schema with a small hybrid part. The star schema approach was selected because it simplifies analysis by connecting fact tables directly to descriptive dimension tables. A small hybrid structure was added using a bridge table to handle the many-to-many relationship between films and actors.

A separate warehouse database called movie_rental_dw was created. The warehouse includes dimension tables such as dim_customer, dim_film, dim_store, dim_staff, dim_date, and dim_actor. It also includes fact tables such as fact_rental, fact_payment, and fact_inventory_snapshot. The SQL schema defines primary keys, foreign keys, and the relationships between the fact and dimension tables.

3.1 Selected Business Processes

Business Process	Description
Film Rental Process	Represents each time a customer rents a film from a store.
Payment Process	Represents payments made by customers for rentals.
Inventory Availability Process	Represents the availability of films in each store at a snapshot point in time.

3.2 Fact Tables

Fact Table	Business Process	Grain	Main Measures
fact_rental	Film rental transactions	One row per rental transaction	rental_count, rental_duration_days, expected_rental_duration_days, late_return_flag, late_days
fact_payment	Customer payment transactions	One row per payment transaction	payment_amount, payment_count
fact_inventory_snapshot	Inventory availability	One row per film per store per snapshot date	inventory_count, rented_count, available_count

The rental fact table stores each rental transaction and connects it to customer, film, store, staff, and date dimensions. The payment fact table stores each payment transaction and is used for revenue analysis. The inventory snapshot fact table stores the number of available and rented copies for each film in each store.

3.3 Dimension Tables

Dimension Table	Main Attributes	Source OLTP Tables	Purpose
dim_date	date_key, full_date, day, month, quarter, year	Dates from rental and payment	Time-based analysis
dim_customer	customer_id, full_name, email, active, address, city, country, store_id	customer, address, city, country	Customer behavior and location analysis
dim_film	film_id, title, description, release_year, rating, rental_rate, language_name, category_name	film, language, film_category, category	Film popularity and revenue analysis
dim_store	store_id, manager_staff_id, address, city, country	store, address, city, country	Store performance analysis
dim_staff	staff_id, full_name, email, username, active, store_id, city, country	staff, address, city, country	Staff performance analysis
dim_actor	actor_id, full_name	actor	Actor-related film analysis
bridge_film_actor	film_key, actor_key	film_actor	Handles many-to-many film-actor relationship

3.4 Shared Dimensions

Dimension	Used By
dim_date	fact_rental, fact_payment, fact_inventory_snapshot
dim_customer	fact_rental, fact_payment
dim_film	fact_rental, fact_payment, fact_inventory_snapshot
dim_store	fact_rental, fact_payment, fact_inventory_snapshot
dim_staff	fact_rental, fact_payment

Using shared dimensions allows consistent reporting across different business processes. For example, revenue and rental activity can both be analyzed by film, store, customer, staff, and time.

4. Dimensional Model Diagram

The dimensional model diagram was generated using MySQL Workbench Reverse Engineer from the implemented movie_rental_dw schema.

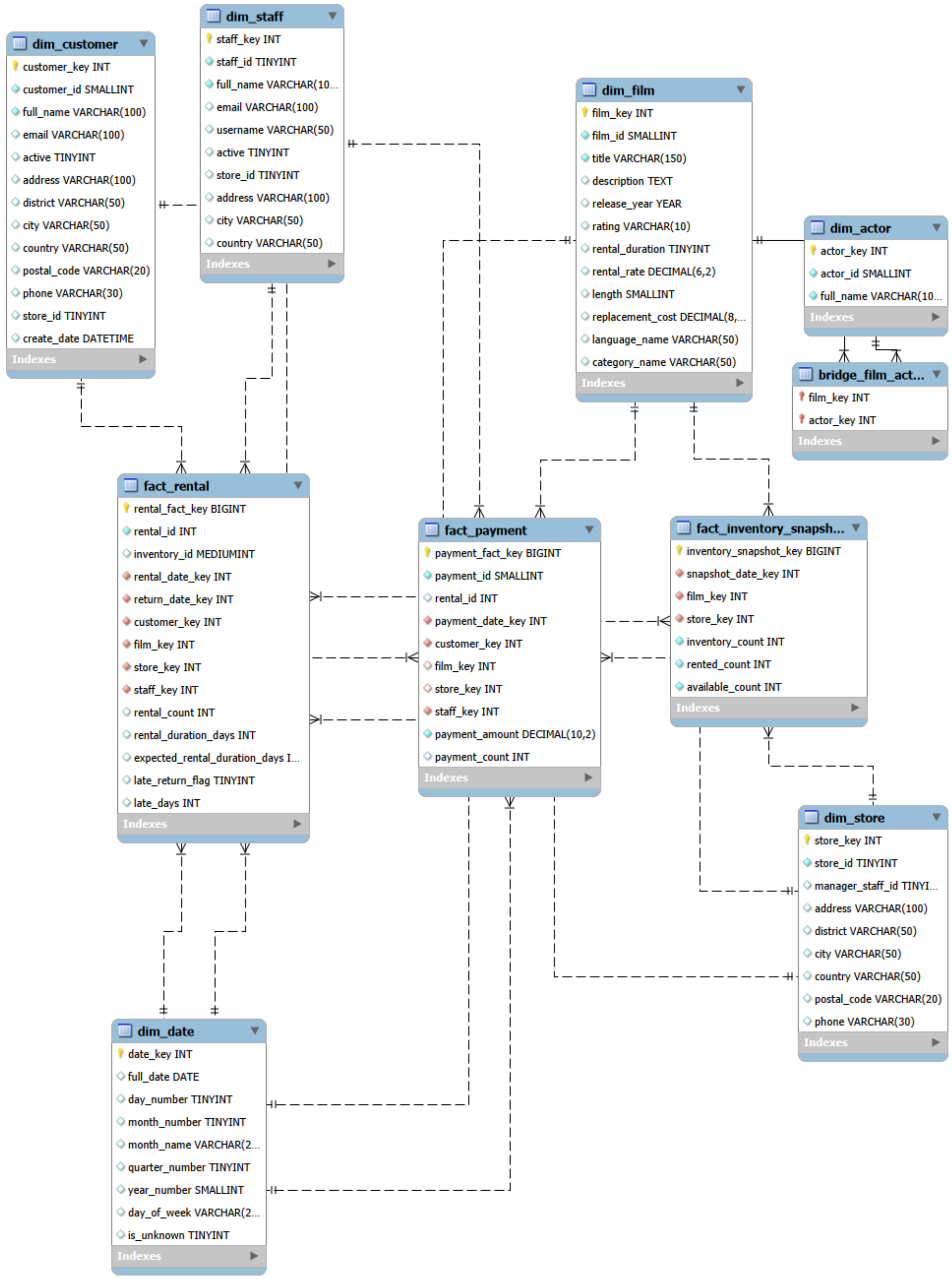

Figure 1: Movie Rental Data Warehouse Dimensional Model

The diagram shows three main fact tables: fact_rental, fact_payment, and fact_inventory_snapshot. These fact tables are connected to shared dimensions such as dim_date, dim_customer, dim_film, dim_store, and dim_staff. The model is mainly a star schema because the fact tables connect directly to dimension tables. The bridge_film_actor table adds a small hybrid part to the design because films and actors have a many-to-many relationship.

5. ETL Design

The ETL process was implemented using SQL scripts. The purpose of ETL is to extract data from the Sakila OLTP database, transform it into analytical structures, and load it into the movie_rental_dw data warehouse. The dimension tables are loaded before the fact tables because fact tables depend on dimension surrogate keys.

5.1 Extract

Source Table(s)	Reason for Extraction
rental	Source for rental transactions and rental dates
payment	Source for payment amounts and payment dates
customer	Source for customer details
film	Source for film details
inventory	Connects films to stores and rentals
store	Source for store information
staff	Source for staff information
address, city, country	Adds address and location information
language, category, film_category	Adds film language and category information
actor, film_actor	Adds actor information and film-actor relationships

5.2 Transform

- Customer data was joined with address, city, and country data to create dim_customer.
- Film data was joined with language, film_category, and category data to create dim_film.
- Store and staff data were enriched with address, city, and country information.
- First name and last name were combined into full_name for customers, staff, and actors.
- Date keys were created from rental dates, return dates, payment dates, and the current inventory snapshot date.
- Rental duration, late return flag, and late days were calculated during fact loading.
- Missing return dates were handled using an unknown date key.
- The film-actor many-to-many relationship was handled using bridge_film_actor.
- Inventory count, rented count, and available count were calculated for each film in each store.

5.3 Load

The loading order was:

9. dim_date
10. dim_customer
11. dim_film
12. dim_store
13. dim_staff
14. dim_actor
15. bridge_film_actor
16. fact_rental
17. fact_payment
18. fact_inventory_snapshot

For example, when loading fact_rental, the ETL script joins the OLTP rental table with inventory, then looks up the related customer, film, store, and staff records in the dimension tables. This allows each rental transaction to be stored with the correct warehouse keys.

6. Analytical Reports

After the data warehouse was loaded, analytical SQL queries were created to answer the business questions. The reports were generated from the movie_rental_dw warehouse, not directly from the OLTP schema.

Report	Purpose
Top 10 Most Rented Films	Identifies films with the highest rental activity
Top 10 Films by Revenue	Identifies films that generated the highest revenue
Revenue by Store	Compares store revenue performance
Monthly Revenue Trend	Shows how revenue changes over time
Monthly Rental Trend	Shows how rental activity changes over time
Most Active Customers	Identifies customers with the highest rental activity and revenue
Staff Performance	Compares staff by processed rentals, processed payments, and revenue
Late Returned Films	Identifies films that are returned late most often

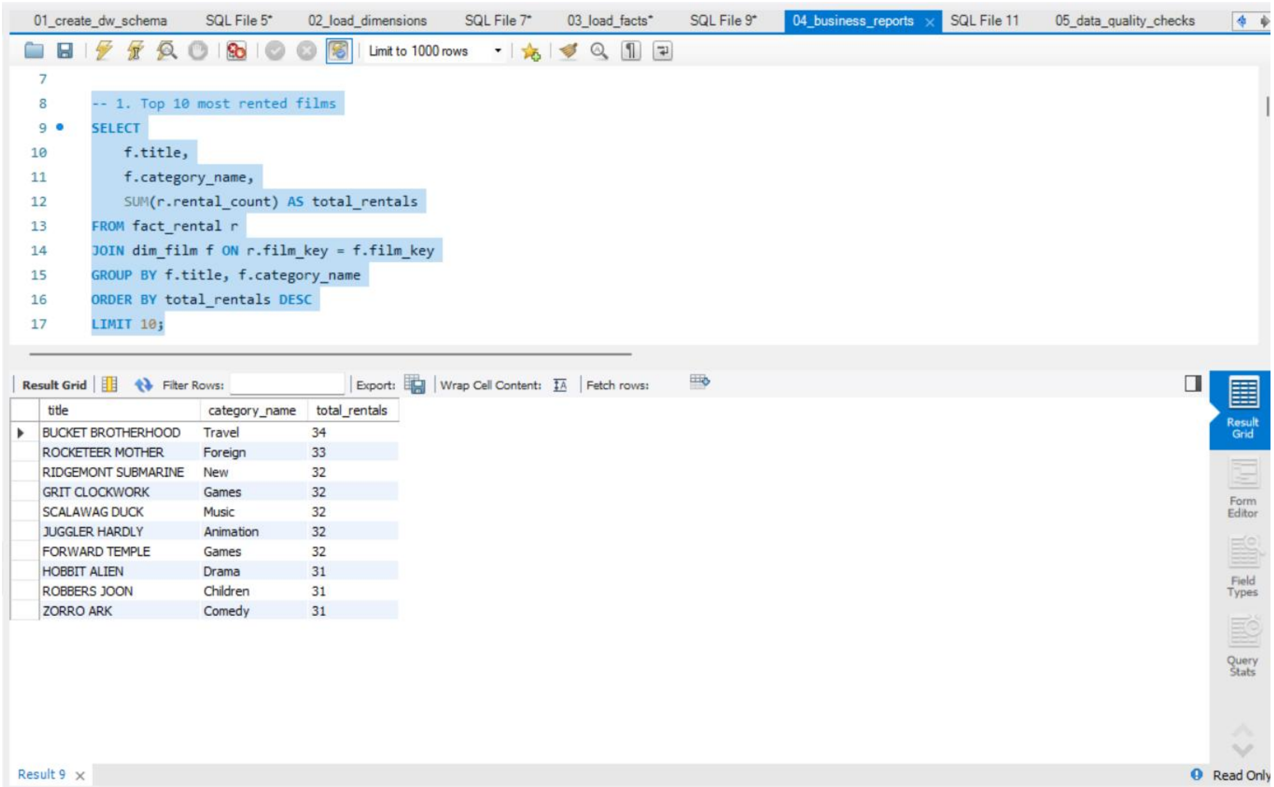

Figure 2: Top 10 Most Rented Films

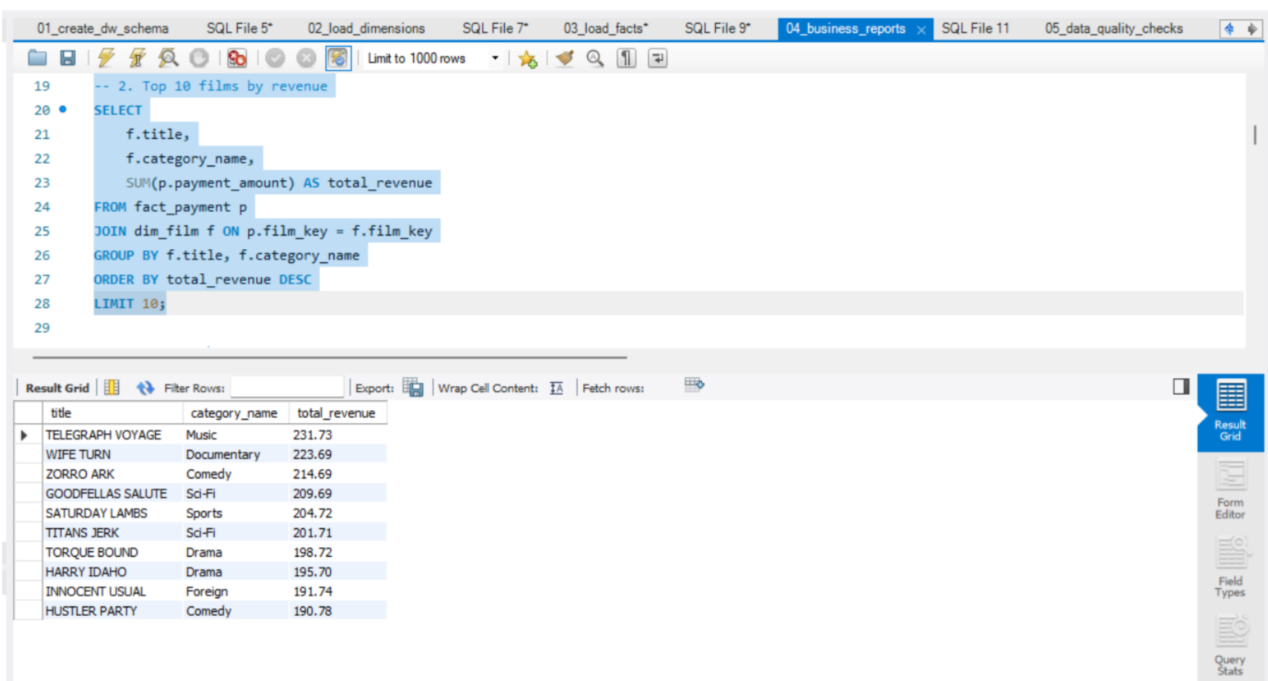

Figure 3: Top 10 Films by Revenue

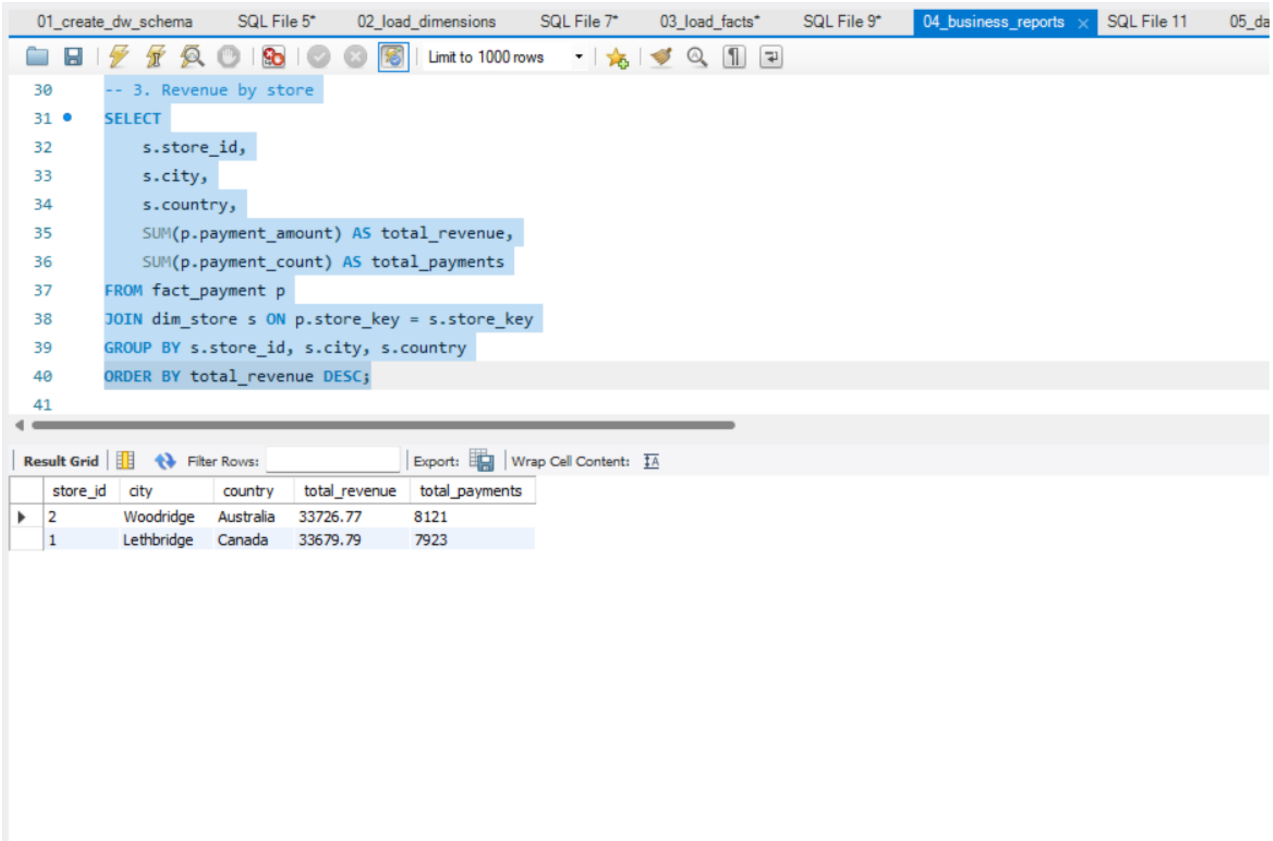

Figure 4: Revenue by Store

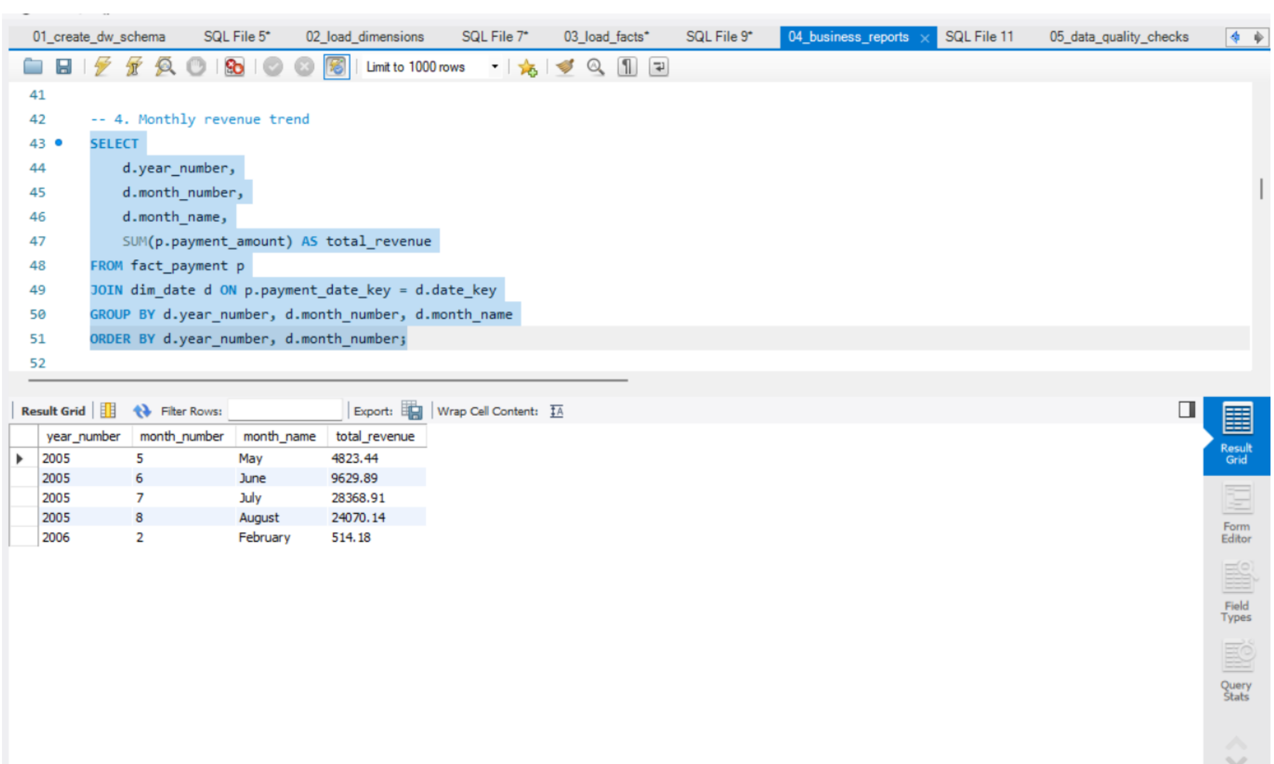

Figure 5: Monthly Revenue Trend

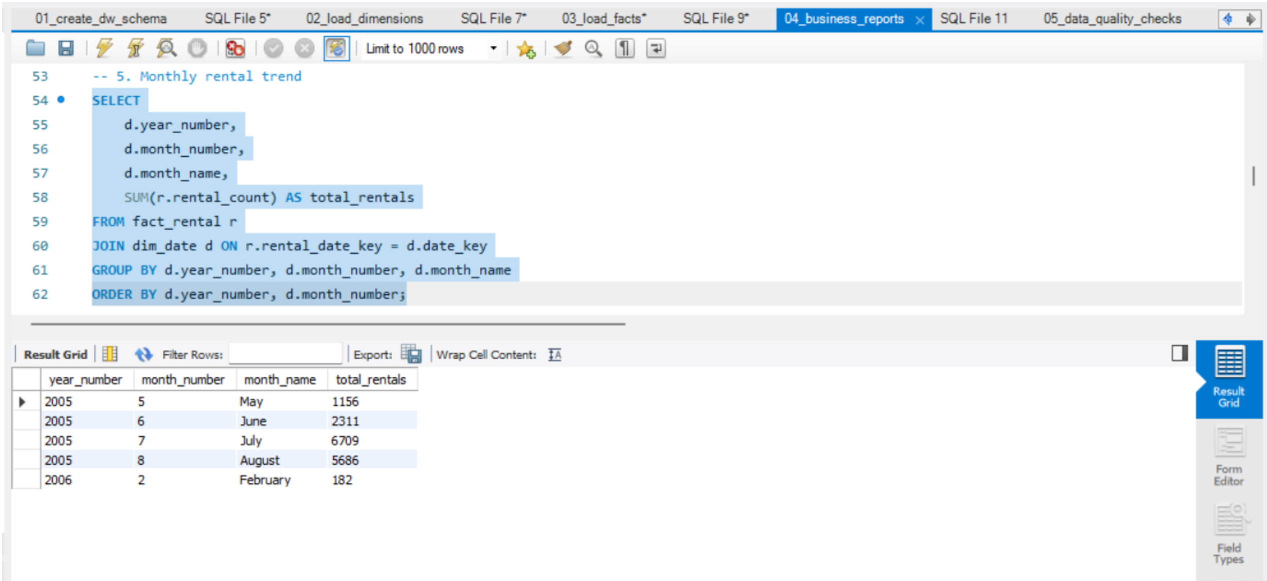

Figure 6: Monthly Rental Trend

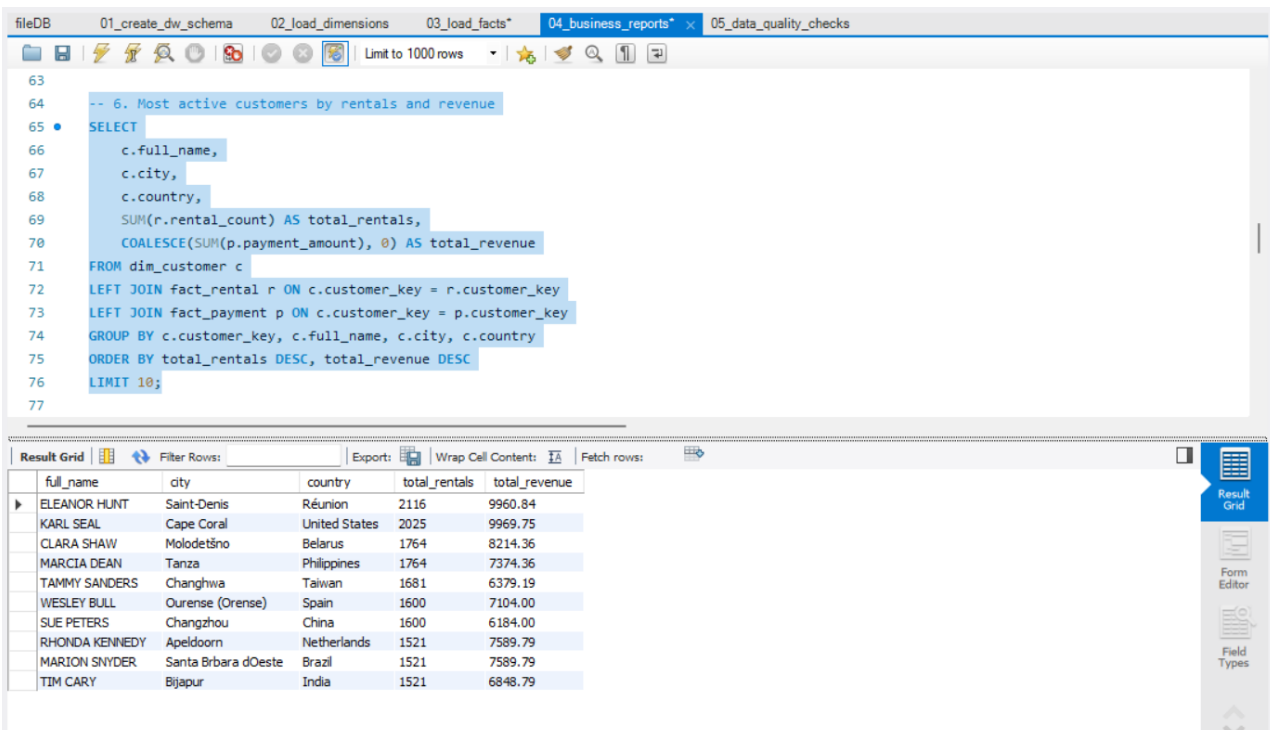

Figure 7: Most Active Customers by Rentals and Revenue

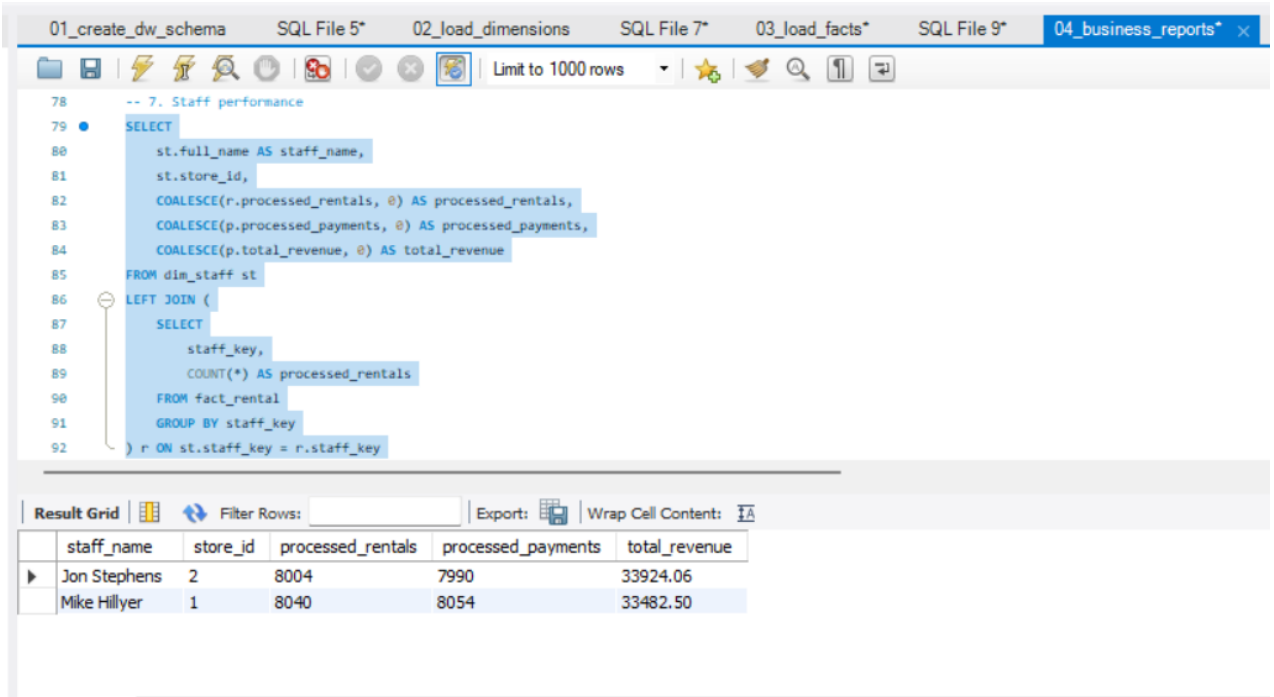

Figure 8: Staff Performance

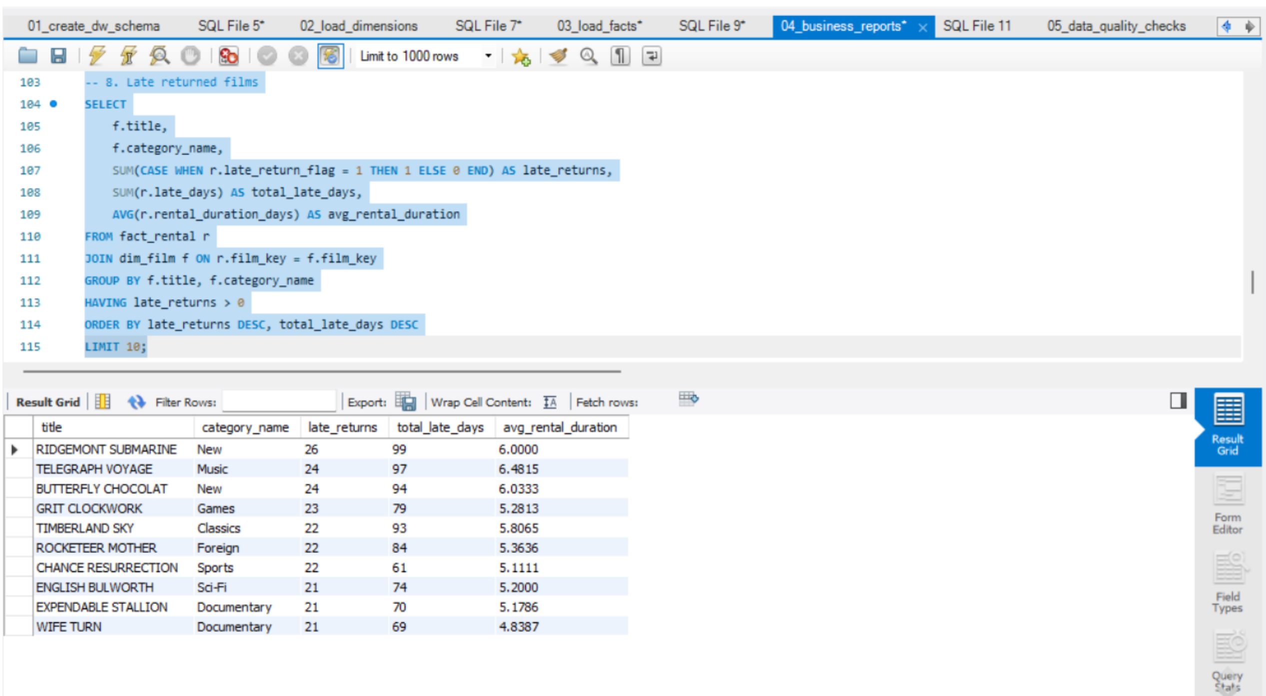

Figure 9: Late Returned Films

7. Data Quality Considerations

Data quality checks were included to make sure that the warehouse was loaded correctly and that the analytical reports are reliable. This section is kept short because the main focus of the project is dimensional modeling and ETL design.

- Check duplicated rental records in fact_rental.
- Check duplicated payment records in fact_payment.

- Check invalid payment amounts where payment_amount <= 0.
- Check cases where the return date is earlier than the rental date.
- Check negative late days.
- Check missing dimension references in fact_rental.
- Count the rows loaded into each warehouse table.

Table Name	Row Count
dim_date	92
dim_customer	599
dim_film	1000
dim_store	2
dim_staff	2
dim_actor	200
bridge_film_actor	5462
fact_rental	16044
fact_payment	16044
fact_inventory_snapshot	1521

These counts confirm that the dimension and fact tables were populated successfully. The checks also help ensure that reports are based on valid and consistent warehouse data.

8. Conclusion

This project transformed the Sakila movie rental OLTP database into a dimensional data warehouse. A separate warehouse database called movie_rental_dw was created to support analytical reporting and business decision-making.

The warehouse design includes three fact tables: fact_rental, fact_payment, and fact_inventory_snapshot. It also includes several dimension tables such as dim_date, dim_customer, dim_film, dim_store, dim_staff, and dim_actor. A bridge table was used to handle the many-to-many relationship between films and actors.

ETL scripts were used to extract data from the OLTP database, transform it into analytical structures, and load it into the warehouse. The final reports demonstrate that the warehouse can answer important business questions related to film popularity, revenue, store performance, customer behavior, staff performance, monthly trends, late returns, and inventory availability.

Overall, the proposed data warehouse improves the ability of business managers to analyze operations and make data-driven decisions.

Appendix A: Chart Visualizations

The following charts were generated using Python, pandas, and matplotlib from the analytical SQL reports. They provide visual summaries of the main business questions, including film popularity, revenue performance, monthly trends, customer activity, staff performance, and late returns.

Chart	Business Question
Top 10 Most Rented Films	Which films are rented most frequently?
Top 10 Films by Revenue	Which films generate the highest revenue?
Revenue by Store	Which stores generate the highest revenue?
Monthly Revenue Trend	How does revenue change by month?
Monthly Rental Trend	How does rental activity change by month?
Most Active Customers	Which customers rent the most films?
Staff Performance	Which staff members process the most activity and revenue?
Late Returned Films	Which films are returned late most often?

Top 10 Most Rented Films

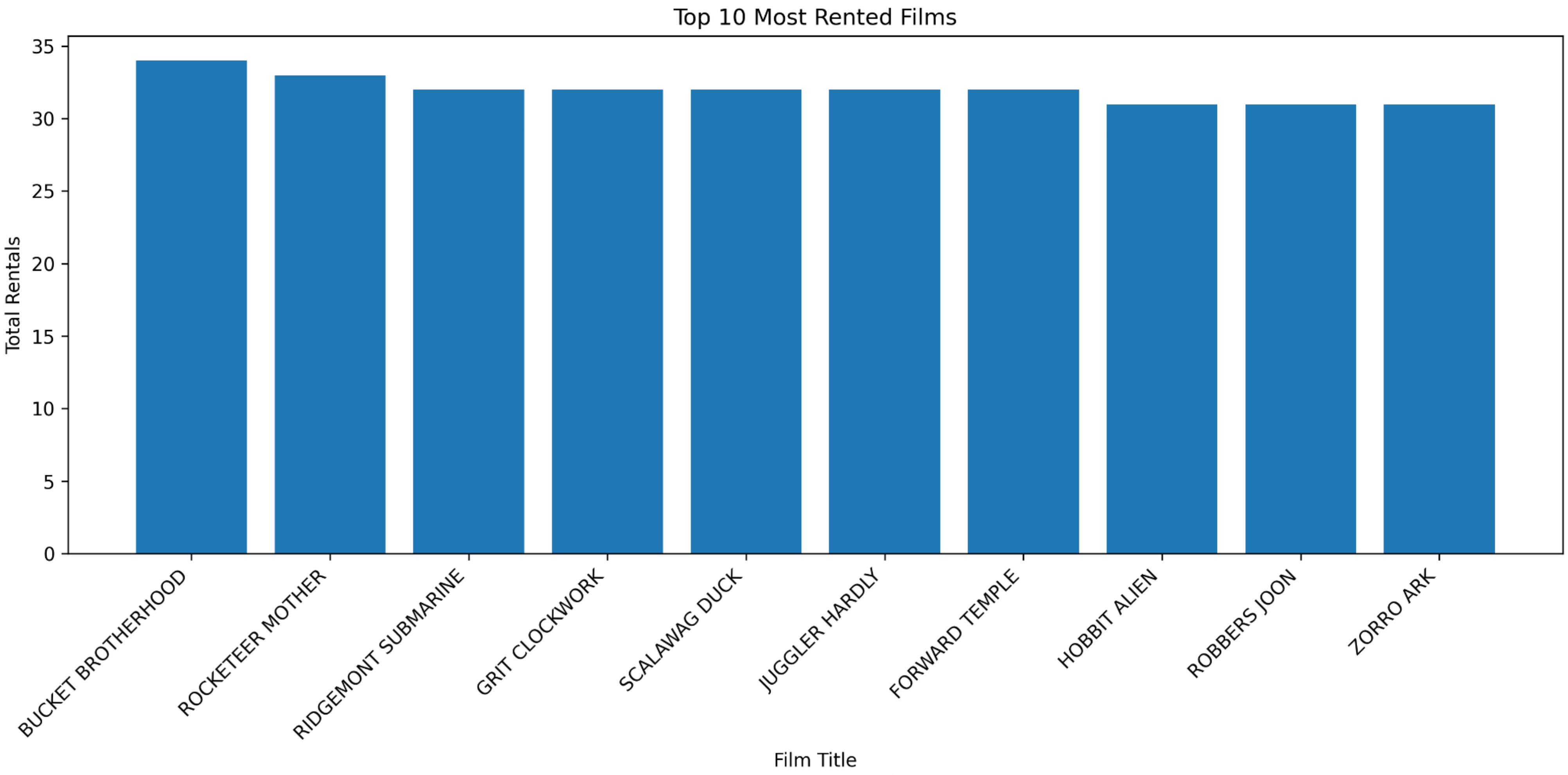

Figure A.1: Shows the films with the highest rental activity. BUCKET BROTHERHOOD is the most rented film, followed by ROCKETEER MOTHER and RIDGEMONT SUBMARINE.

Top 10 Films by Revenue

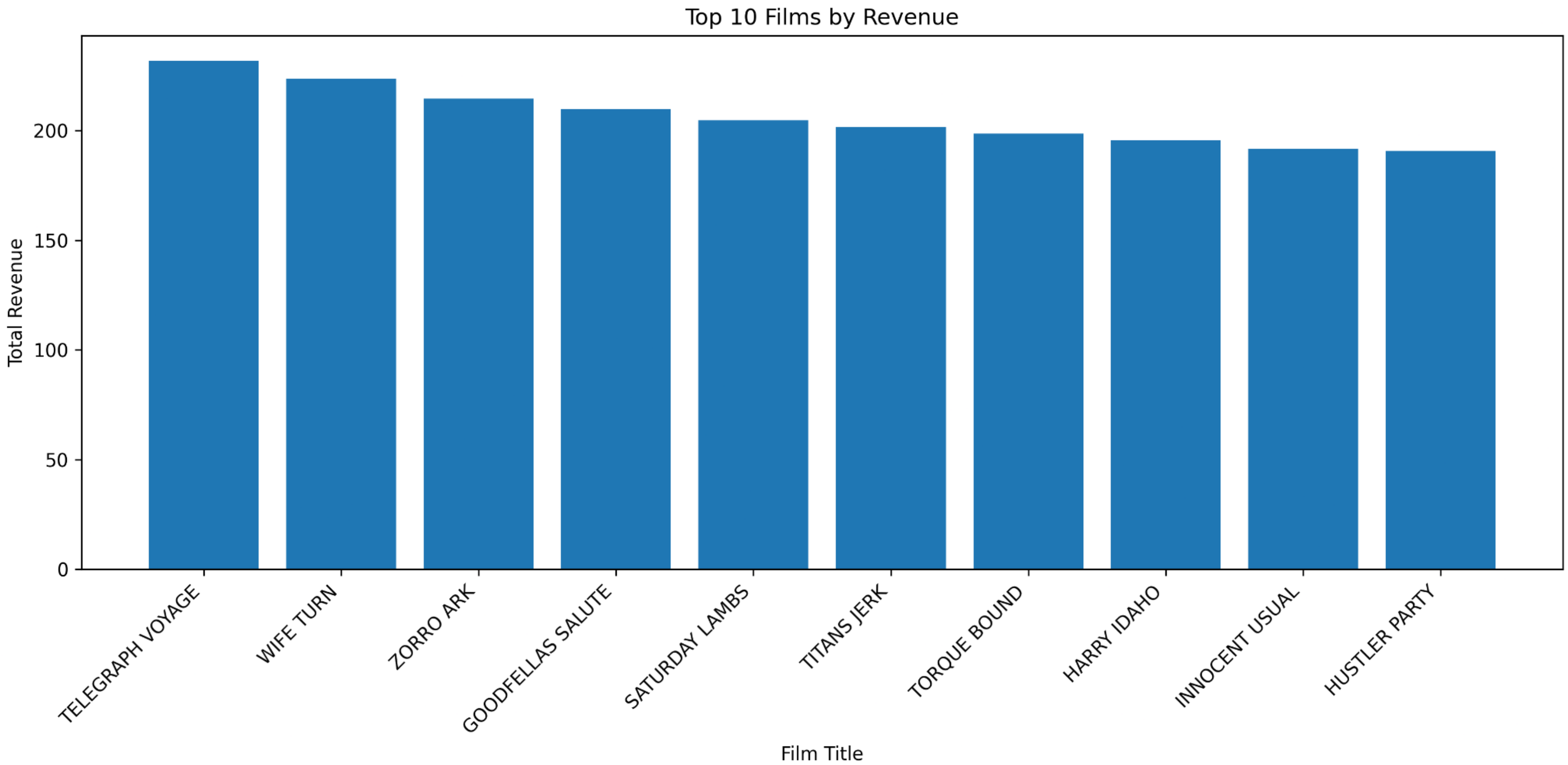

Figure A.2: Shows the films that generated the highest total revenue. TELEGRAPH VOYAGE generated the highest revenue, followed by WIFE TURN and ZORRO ARK.

Revenue by Store

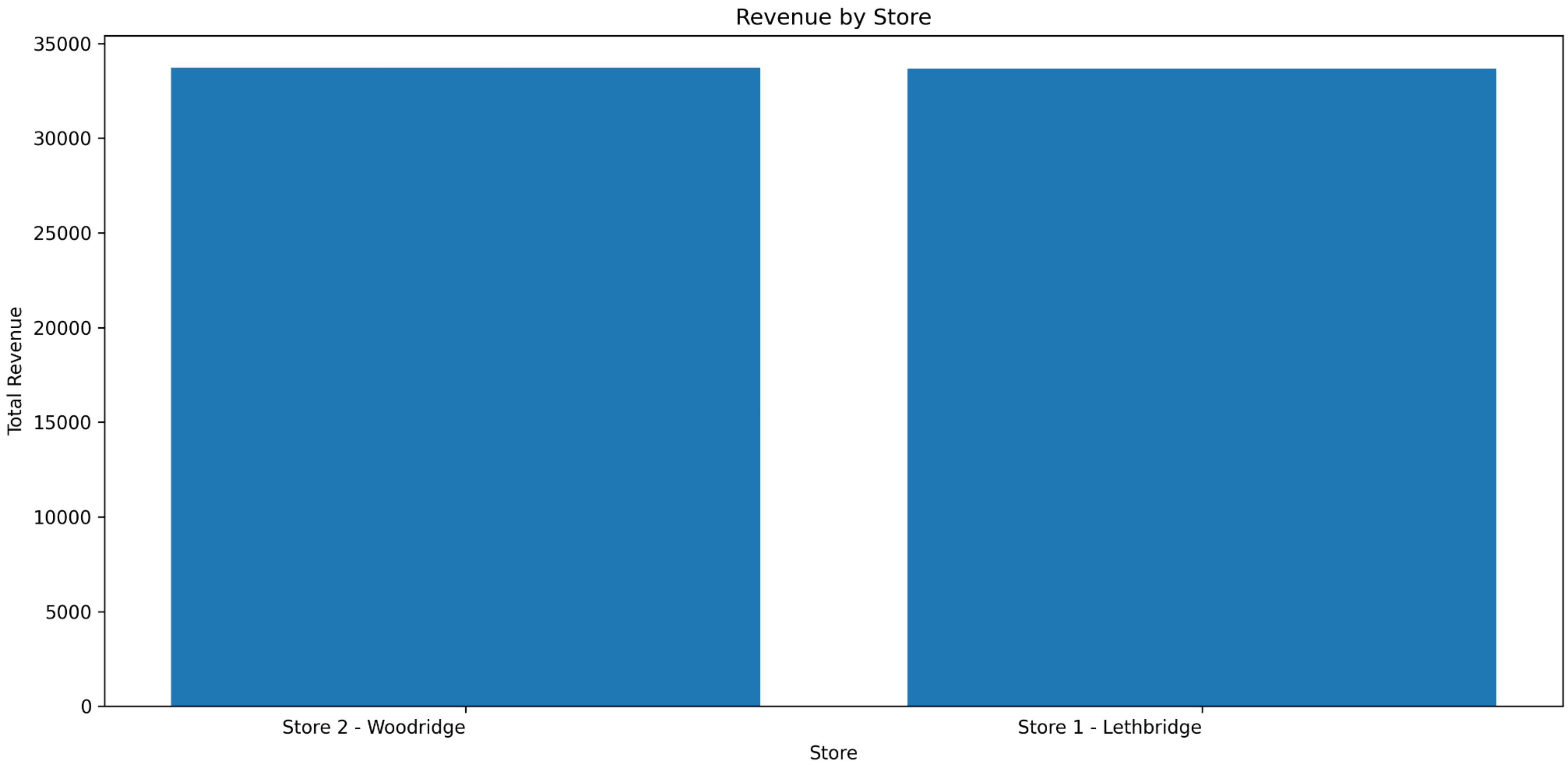

Figure A.3: Compares total revenue across stores. Store 2 - Woodridge and Store 1 - Lethbridge show very close revenue performance.

Monthly Revenue Trend

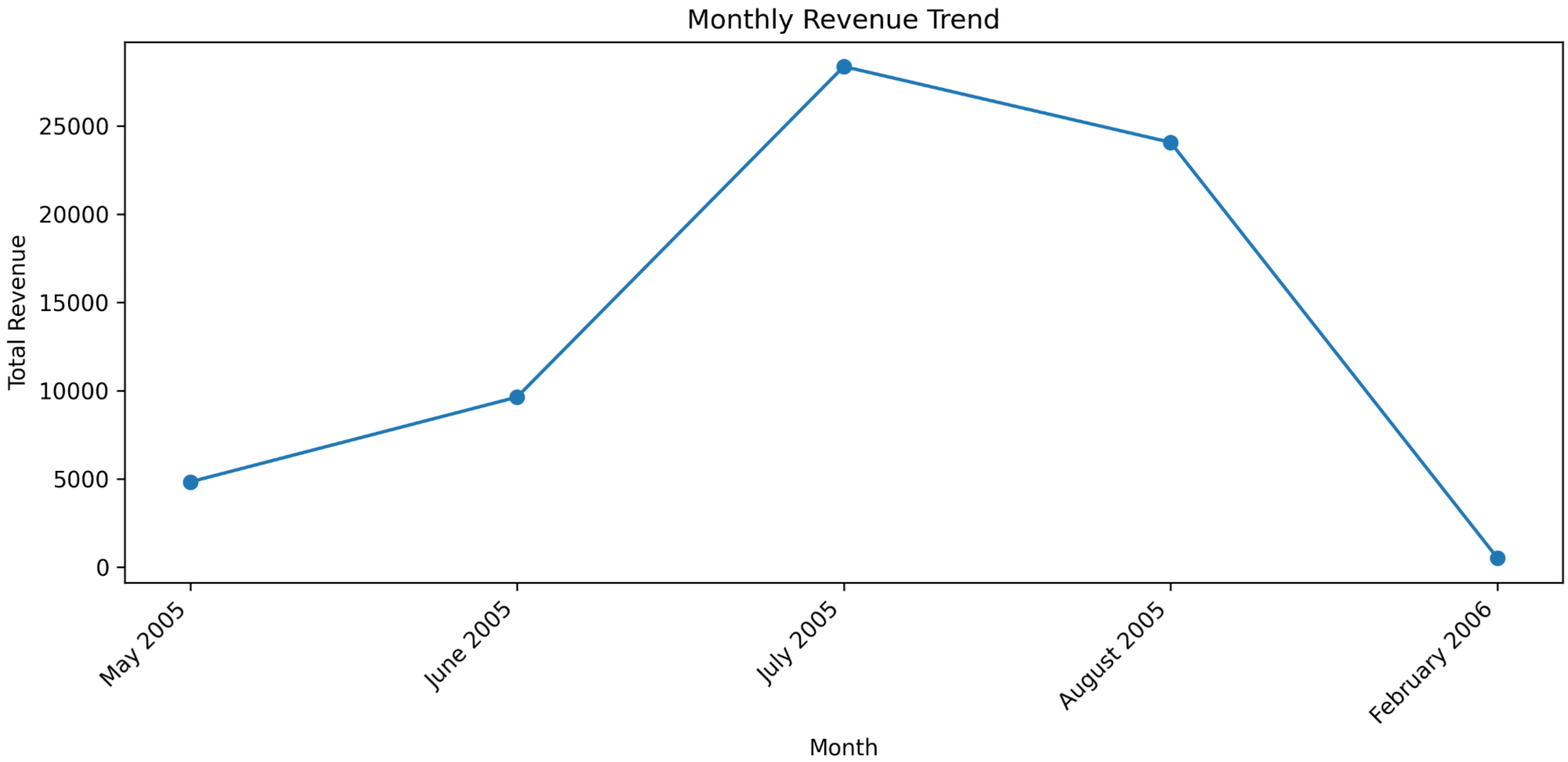

Figure A.4: Shows revenue over time. Revenue increased from May to July 2005, peaked in July, and then decreased after August.

Monthly Rental Trend

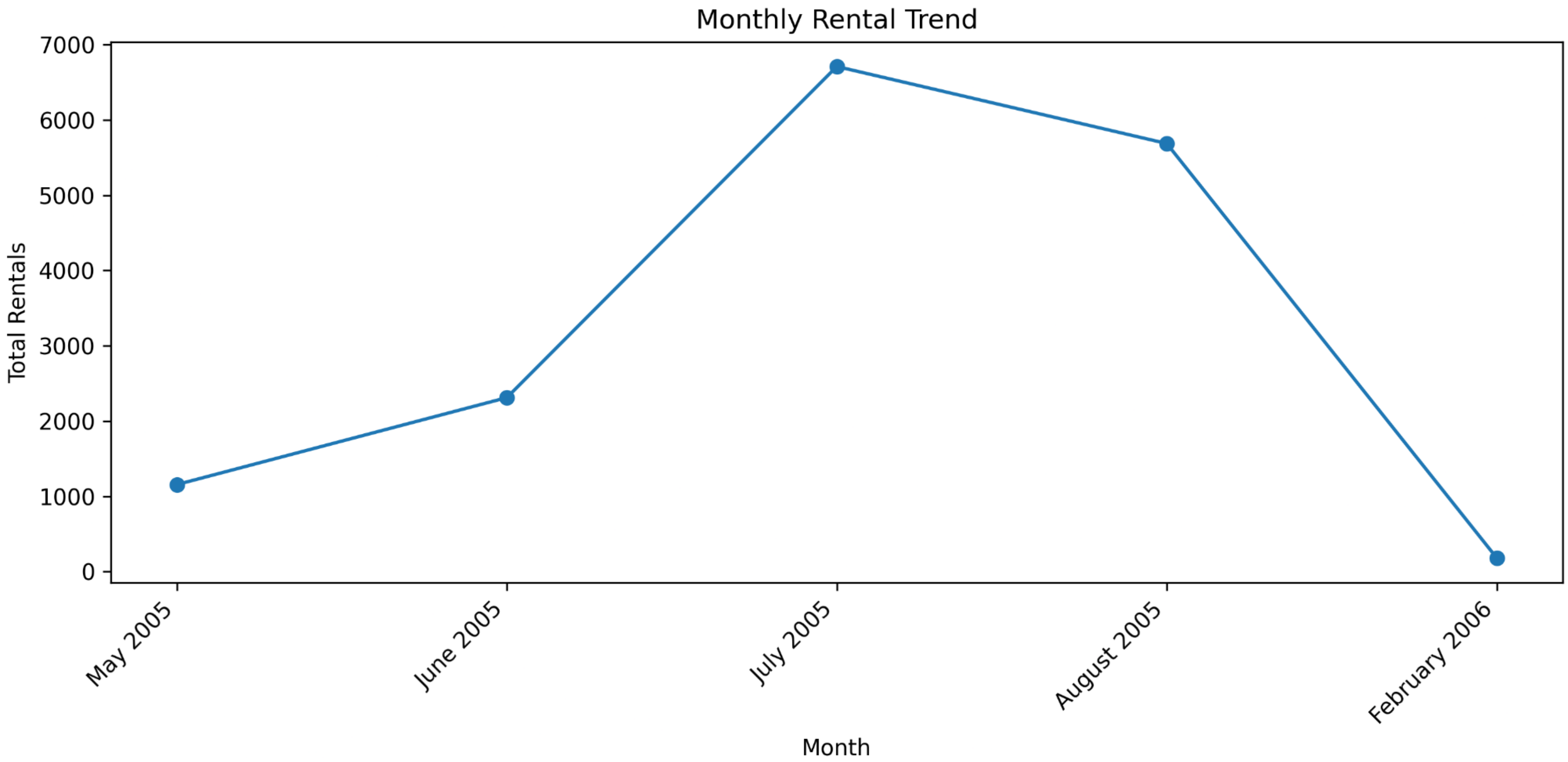

Figure A.5: Shows rental activity over time. Rental activity follows a similar pattern to revenue, with July 2005 as the busiest month.

Most Active Customers by Rentals

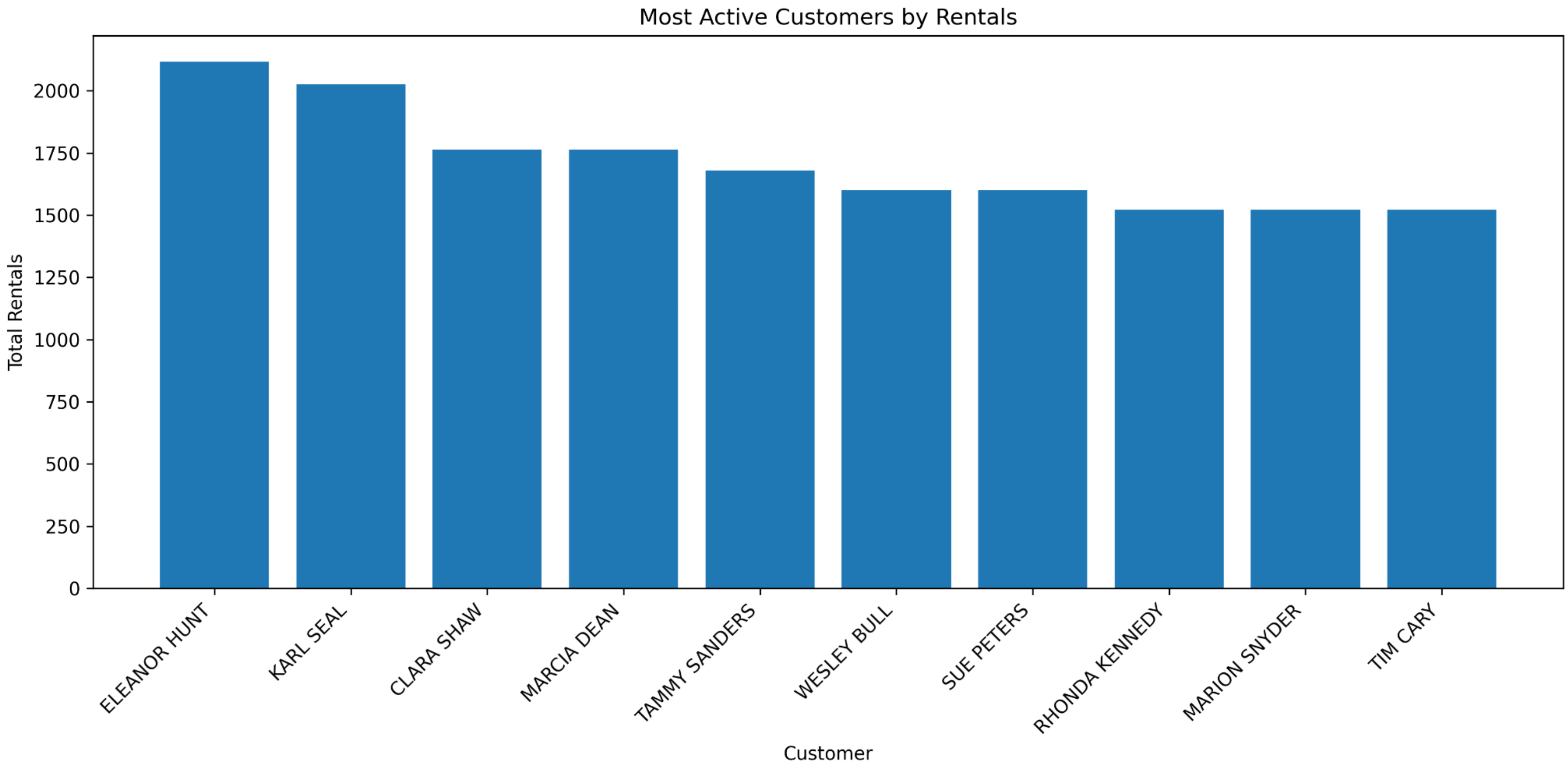

Figure A.6: Shows the customers with the highest rental activity. ELEANOR HUNT is the most active customer, followed by KARL SEAL.

Staff Performance by Revenue

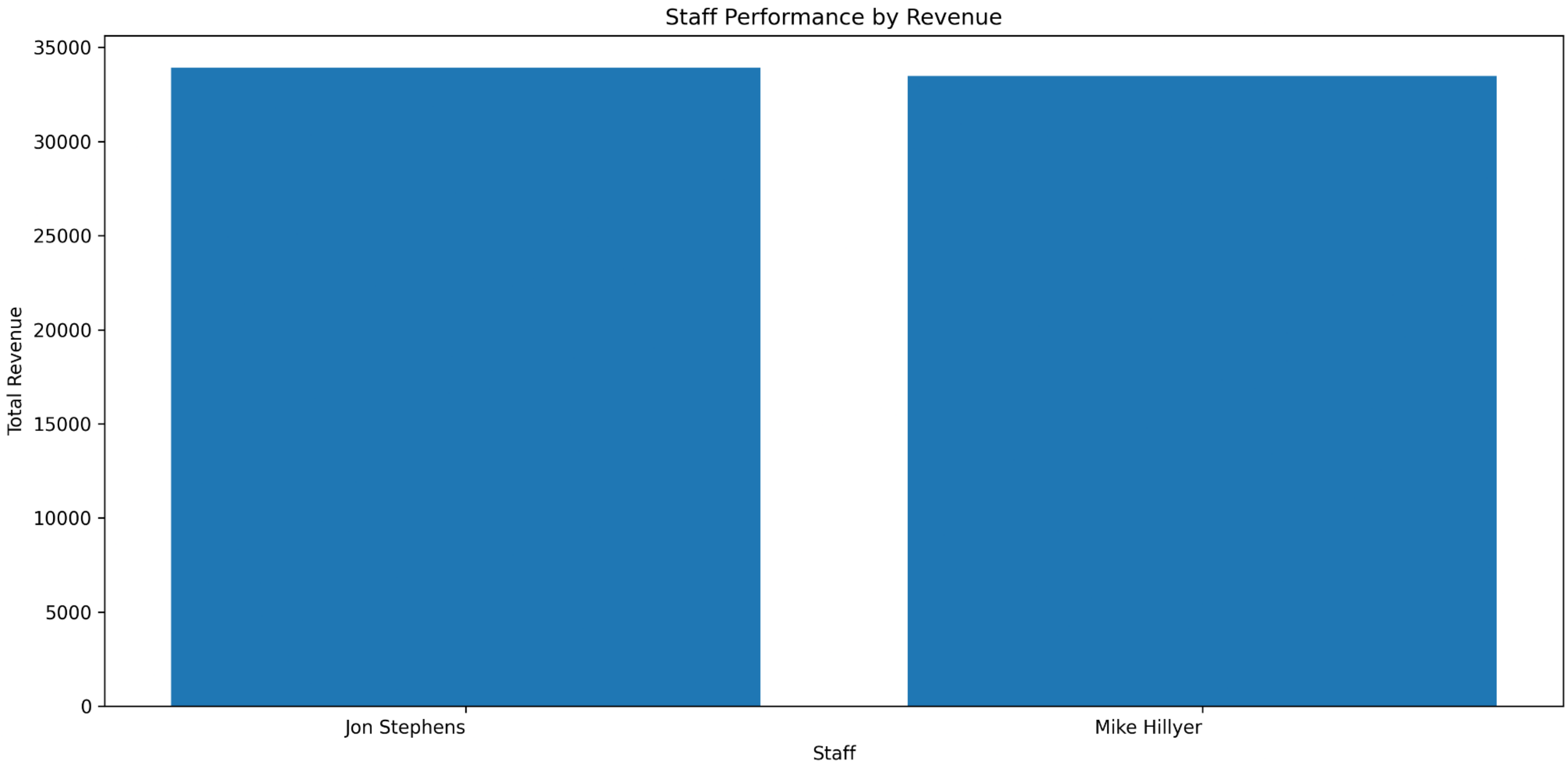

Figure A.7: Compares staff performance by revenue processed. Jon Stephens has slightly higher processed revenue than Mike Hillyer.

Late Returned Films

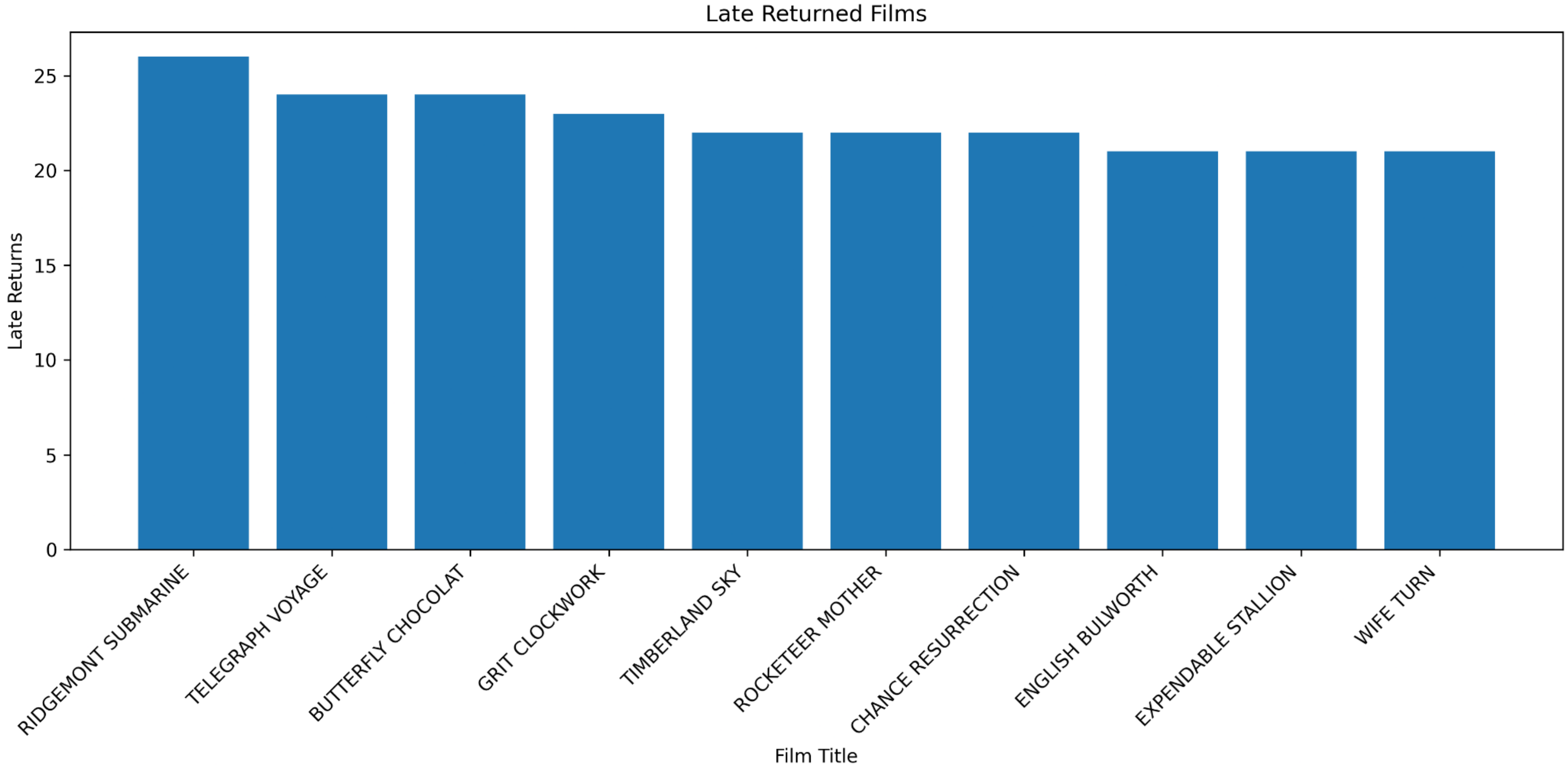

Figure A.8: Shows films returned late most often. RIDGEMONT SUBMARINE has the highest number of late returns.